Frontier Gaps as an Interface Between Clearance Reasoning and Contact-Feasible Multi-Robot Pushing

***Abstract*—Cluttered path clearing is not only about detecting blockages, but about deciding which width-constrained bottlenecks are worth intervening on and whether they can be opened through executable multi-robot interactions. Existing Navigation Among Movable Obstacles (NAMO) methods mainly reason about obstacle rearrangement at an abstract geometric level, while collaborative pushing methods focus on local contact execution without explicit guidance for multi-obstacle corridor construction. We present frontier gaps as a structured interface that connects these levels by converting reachable clearance bottlenecks into contact-feasible intervention queries. A W-clearance connectivity graph (WCCG) determines whether a traversable corridor already exists and, when it does not, exposes the reachable frontier gaps that currently block progress. For each selected gap, only gap-conditioned pushing candidates are generated for the adjacent movable obstacles, and short-horizon physical simulation is used to validate these induced interventions rather than to support generic downstream search. Building on this interface, a physics-informed hybrid search prioritizes and selectively expands only the most promising candidates. This formulation links bottleneck selection to executable contact generation, producing corridor-opening actions grounded in both scene geometry and physical feasibility. Simulation results and hardware experiments show that frontier-gap-guided planning improves path-clearing reliability while yielding action sequences that remain consistent with real multi-robot execution.**

I. Introduction

Large transport vehicles often need to traverse cluttered environments in which direct passage is blocked by movable obstacles such as pallets, boxes, or equipment. In these settings, planning must go beyond collision-free routing: the environment itself must be reconfigured to create a corridor with sufficient clearance [1], [2]. This challenge arises in warehouse automation, disaster response, and other dense workspaces where obstacle interaction is unavoidable. The resulting problem is therefore not only to detect whether passage is blocked, but also to decide how the blockage can be removed through physical interaction.

The difficulty becomes sharper when corridor construction is delegated to a team of smaller mobile robots acting on behalf of a larger transport platform [3], [4]. The large vehicle defines the required clearance but does not participate in pushing, so the robot team must identify and execute coordinated contacts that open a traversable corridor for downstream passage. Cooperative pushing is therefore a natural mechanism for path clearing, but it also makes feasibility strongly contact-dependent: pushing one obstacle can induce coupled motion of nearby objects, and whether a candidate action succeeds depends on local geometry, robot size, and obstacle arrangement [5], [6]. As a result, a geometrically plausible clearing sequence may still fail if the required contacts and motions are not executable.

Existing methods typically capture only part of this problem. NAMO-style approaches reason about obstacle rearrangement and corridor creation, but often abstract away the contact mechanics that determine whether an intervention can actually be executed in tightly cluttered scenes [2], [7]. In contrast, collaborative pushing and multi-robot manipulation methods focus on coordinated contact and force generation, but usually consider single-object transport or more structured settings [8], [9]. Physics-informed planning adds simulation-based validation, yet this validation is often only loosely coupled to the upstream decision of which blockage should be addressed first [10], [11]. For cluttered path clearing, the missing link is therefore a representation that connects scene-level bottleneck selection to executable multi-robot contact planning.

We address this disconnect through *frontier gaps*, which serve as a structured interface between scene-level clearance reasoning and contact-feasible multi-robot pushing. Concretely, a width-aware W-clearance connectivity graph (WCCG) identifies whether a traversable corridor already exists and, when it does not, exposes the reachable frontier gaps that currently block progress. Each selected gap is then converted into a compact set of gap-conditioned pushing candidates for the adjacent movable obstacles, so that physical simulation is used only to selectively validate these induced interventions. This interface lets cluttered path clearing be treated as a coupled planning problem that links bottleneck selection to executable contact generation for multi-robot corridor construction.

II. Method

The proposed method is organized around a structured interface between scene-level clearance reasoning and contact-feasible multi-robot execution. Given a start–goal query for a large vehicle in a cluttered environment, it first uses a width-aware representation to test whether a W-clear corridor already exists. If passage is blocked, the representation returns a small set of reachable *frontier gaps* that define the current bottlenecks for progress. These bottlenecks are then converted into compact multi-robot intervention queries and selectively validated through physics-informed search. The overall pipeline is shown in Fig. 1.

A. Clearance Representation and Frontier Gaps

We represent the scene with a *W-clearance connectivity graph* (WCCG), whose role is not to recover a detailed geometric path at every step, but to expose the actionable output needed for path clearing. The graph is constructed directly from obstacle geometry in a width-aware manner. In particular, it uses centroid nodes to represent obstacle components and bridge relations to represent local inter-obstacle passages. The resulting graph is

$$\mathcal{G}_W \triangleq (\mathcal{V}, \mathcal{E}_c \cup \mathcal{E}_b), \tag{1}$$

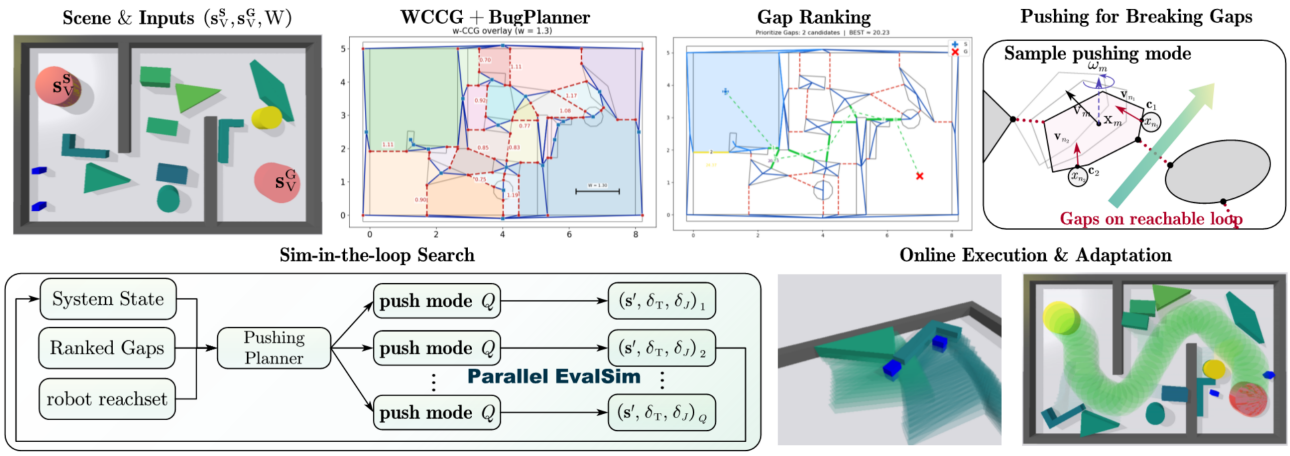

Fig. 1. Overview of the proposed interface-centered formulation. **Top:** The WCCG identifies reachable frontier gaps and induces gap-conditioned pushing candidates. **Bottom-left:** Physics-informed hybrid search. **Bottom-right:** Online execution.

where $\mathcal{V}$ contains centroid and bridge nodes, $\mathcal{E}_c$ denotes centroid–bridge edges, and $\mathcal{E}_b$ denotes bridge–bridge edges. Each bridge edge is annotated with a local width w_{uv}, defined by the distance between the associated closest points on the two neighboring components. This makes $\mathcal{G}_W$ a compact width-aware representation of where the scene is already traversable and where narrow passages arise.

Given the vehicle start and goal, we perform a width-constrained connectivity query on $\mathcal{G}_W$. If the two lie in the same W-clear connected face, then a feasible corridor already exists and no clearing action is needed. Otherwise, the query returns the current reachable *frontier loop* $\mathcal{L}$ that separates the two sides under the width constraint. The bridge–bridge edges visible on $\mathcal{L}$ define the first-hop set of reachable bottlenecks,

$$\Gamma_{\mathcal{L}} \triangleq \{g_1, \ldots, g_K\}, \tag{2}$$

where each g_k is a candidate *frontier gap* exposed by the current clearance query.

These frontier gaps are the key interface produced by the representation. They are not arbitrary narrow passages or generic local heuristics, but the first reachable bottlenecks identified by the width-aware connectivity test. As a result, subsequent stages do not generate contact actions over the full scene. Instead, they operate only on $\Gamma_{\mathcal{L}}$, converting these scene-level clearance bottlenecks into compact contact-feasible intervention queries for multi-robot pushing.

B. Contact-Feasible Pushing Candidates

For a frontier gap $g \in \Gamma_L$ under consideration, the next question is whether this reachable bottleneck can be enlarged through executable multi-robot interaction. The role of this stage is therefore not to solve a fully general manipulation problem, but to convert a scene-level clearance bottleneck into a small set of *contact-feasible intervention queries*.

For the movable obstacles adjacent to g, we sample a small set of short-horizon obstacle motion intents,

$$\mathbf{v} \in \mathbb{R}^3, \tag{3}$$

biased toward enlarging the target gap while allowing limited rotational components to account for local geometry. For each sampled intent, we instantiate an associated pushing mode

$$\boldsymbol{\xi} \triangleq (\mathcal{C}, \mathbf{u}), \tag{4}$$

where $\mathcal{C}$ denotes the robot–object contact configuration and $\mathbf{u}$ the associated contact wrench parameters. Each retained candidate therefore takes the form $(g, \mathbf{v}, \boldsymbol{\xi})$, and the candidate set induced by gap g is denoted by Ξ_g.

This candidate generation is deliberately gap-conditioned. Instead of searching over arbitrary contact actions in the full scene, the method only considers obstacle motions and contact configurations that are relevant to the selected frontier gap. For each sampled intent, the corresponding mode is generated by a sparse constrained optimization over reachable boundary contacts, under local friction and wrench feasibility constraints. In this way, each candidate mode is explicitly tied to a desired gap-enlarging obstacle motion rather than being hand-crafted or sampled without task context. In implementation, efficiency is further improved through a persistent *ModeTable*, which caches effective mode entries and reuses them across obstacles with similar local geometry.

A candidate is retained only if it passes short-horizon physical validation. Specifically, the robots must be able to reach the proposed pre-contact poses, avoid collision during approach and pushing, maintain stable contact, and induce obstacle motion that actually enlarges the target gap. This validation is performed through a short-horizon rollout,

$$(\mathbf{s}', \delta_{\mathrm{T}}, \delta_{\mathrm{J}}) \triangleq \mathsf{EvalSim}\big(\mathbf{s}, (g, \mathbf{v}, \boldsymbol{\xi})\big), \tag{5}$$

which returns the successor state, time cost, and control effort. As a result, this stage does not merely propose plausible pushes; it filters the reachable frontier gaps into a compact set of physically testable multi-robot interventions for the hybrid search.

C. Selective Validation in Hybrid Search

The representation and contact layers above are coupled through a *physics-informed hybrid search* (PIHS). Rather than exhaustively simulating all obstacles or all contact actions, PIHS incrementally validates only those interventions induced

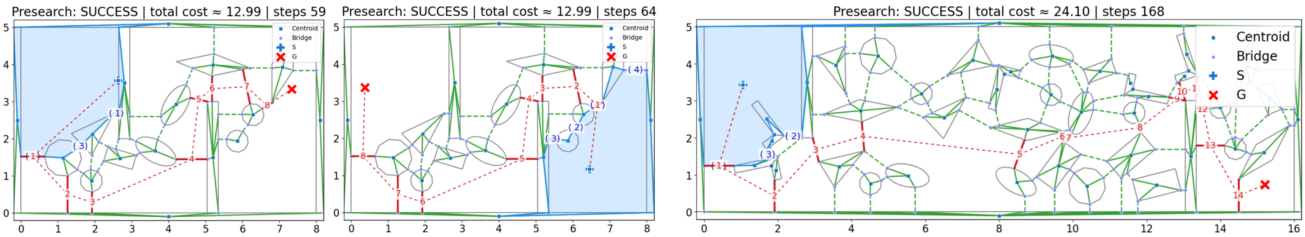

Fig. 2. Illustration of frontier-gap ranking. Each panel shows the WCCG with the current reachable face (light blue). From the frontier loop, candidate bridge–bridge gaps are enumerated, assigned local first-hop priorities (blue), and refined by short presearch over downstream gap-crossing sequences (red). **Left**: swapping start and goal in the same map produces symmetric ranked sequences. **Right**: the same ranking procedure scales to a denser scene with about 30 obstacles.

by the currently reachable frontier gaps. Each search node is defined as

$$\nu \triangleq (s, \pi), \tag{6}$$

where s is the current system state and π stores the partial sequence of previously validated gap-opening actions.

At a given node, the method first updates the WCCG and checks whether a W-clear connection from start to goal already exists. If so, the search terminates and returns the schedule stored in π. Otherwise, the current state induces a reachable frontier loop $\mathcal{L}$ and the associated candidate frontier gaps $\Gamma_{\mathcal{L}}$. These gaps are ranked by a lightweight geometric score,

$$\mathrm{score}(g) = C_{\mathrm{trans}}(g) + C_{\mathrm{wide}}(g) + C_{\mathrm{pre}}(g), \tag{7}$$

where $C_{\mathrm{trans}}(g)$ denotes the robots' transition cost to the gap, $C_{\mathrm{wide}}(g)$ the estimated effort required to widen it, and $C_{\mathrm{pre}}(g)$ a short presearch estimate of downstream gap-crossing cost toward the goal. The node priority is then defined by

$$f(\nu) = \chi(\nu) + \min_{g \in \Gamma_{\mathcal{L}}} \mathrm{score}(g), \tag{8}$$

where $\chi(\nu)$ is the cumulative realized cost of the partial schedule. This ranking acts as a geometric filter before any contact-level simulation is invoked, so that validation is focused on the most promising reachable bottlenecks.

For each highly ranked gap g, the method calls the gap-conditioned candidate generator from Sec. II-B to obtain a compact candidate set Ξ_g. Each candidate action $\tau \triangleq (g, \mathbf{v}, \boldsymbol{\xi})$ is then evaluated by the short-horizon rollout defined in Sec. II-B. Only candidates that remain collision-free, maintain valid contact, and effectively enlarge the target gap are accepted. Successful rollouts produce validated successor states, append the corresponding action to the partial schedule, and are inserted for further expansion. Candidate batches can also be validated in parallel.

PIHS is therefore selective in three senses: the WCCG restricts attention to reachable bottlenecks, gap ranking prioritizes which bottlenecks to test first, and gap-conditioned candidate generation together with short-horizon simulation validates only those interventions worth expanding. Repeating this loop incrementally yields a corridor-construction schedule that specifies which gaps to open and how to open them. Once a validated successor restores W-clear connectivity between the start and goal, the accumulated schedule π defines the final solution.

TABLE I
PERFORMANCE COMPARISON AND ABLATIONS.

Method / Variant	Succ.(%)	PT (s)	ET (s)	#Sims	#Pushes
DFS-WCCG	25.0	>100.0	51.2	>500	8.0
SL-Push	75.0	18.2	64.6	20.0	10.0
Rec-NAMO	37.5	13.3	42.4	0.0	7.0
FG-Interface (ours)	**92.5**	**10.3**	**28.6**	**121.5**	**6.0**
w/o presearch	85.0	16.9	35.2	178	7.0
w/o ModeTable	80.0	15.1	33.4	164	6.0

III. Results and Discussion

We evaluate the proposed formulation in cluttered multi-obstacle clearing tasks with three objectives: to compare its performance against representative baselines, to assess the contribution of frontier-gap-guided prioritization and structured contact generation through ablations, and to verify that the validated interventions remain executable on hardware.

A. Quantitative Evidence

All numerical experiments are implemented in Python3 and PyBullet[12]. The simulation uses a step of $\Delta t = 1/240\,\mathrm{s}$ and a control period of $1/40\,\mathrm{s}$. The nominal scenario is an $8 \times 5\,\mathrm{m}$ bounded workspace in which two robots must clear a passage for the external vehicle through scenes containing 10–15 movable obstacles placed randomly under a minimum separation. Movable obstacles have masses uniformly sampled from $[5, 15]\,\mathrm{kg}$, while immovable structures are modeled with zero mass. A trial is counted as successful when a W-clear path exists from start to goal and the target reaches its goal disk. For fairness, all baselines are evaluated under the same W-clearance criterion and contact model; for the proposed method, each task rollout uses a horizon of 80 simulation steps, and up to 64 candidate push tasks are generated per expansion.

Table I summarizes the main quantitative results. We compare the proposed method against three representative baselines: *DFS-WCCG*, a simulation-in-the-loop depth-first search guided by the WCCG with four fixed axis-aligned pushes for each movable obstacle; *SL-Push*, which follows a straight or waypointed corridor from start to goal, clears the obstacles intersecting that route, and then validates those route-induced pushes with physics; and *Rec-NAMO*, a representative traditional NAMO baseline that incrementally improves local traversability but often fails to establish a complete W-clear corridor for the large vehicle. Across the evaluated scenes, the proposed method achieves the highest success rate at 92.5% while also maintaining the lowest combined

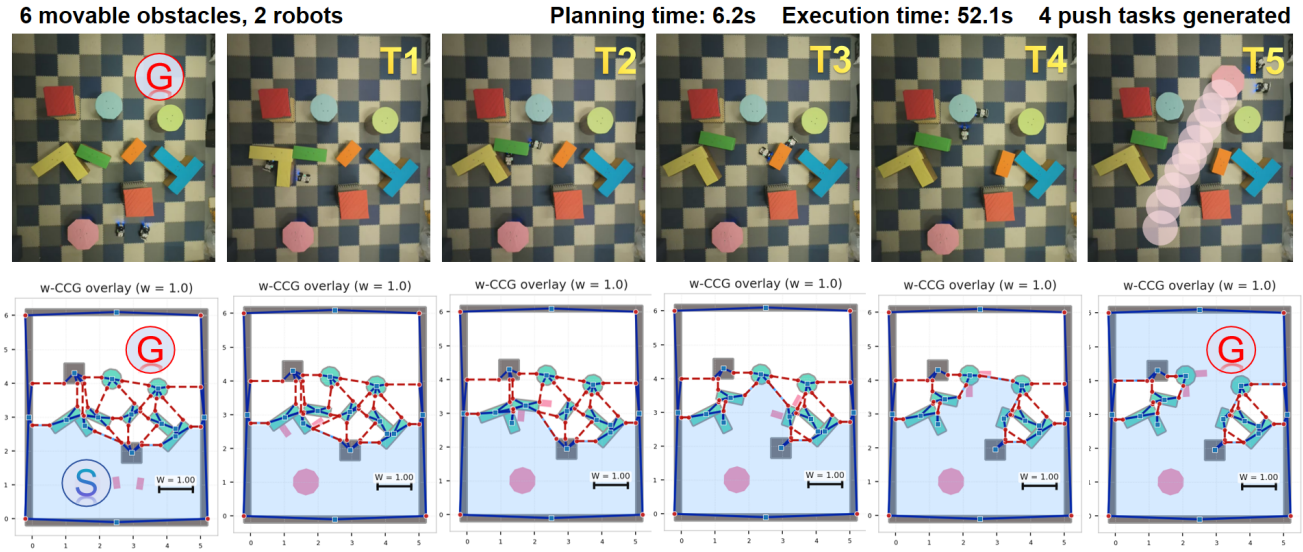

Fig. 3. Real-world pushing experiment with execution–planning alignment. **Top:** snapshots of two robots sequentially opening gaps in a cluttered workspace. **Bottom:** corresponding WCCG overlays, with the current reachable face shown in blue. As the scene evolves, the updated graph remains aligned with the observed object motions and supports continued frontier-gap-guided intervention.

planning and execution time among the successful planners. In contrast, *DFS-WCCG* succeeds in only 25.0% of cases because graph-guided branching quickly becomes intractable. *SL-Push* improves physical feasibility relative to route-only clearing, but it requires many simulation calls and longer action sequences. *Rec-NAMO* is faster than simulation-heavy alternatives, yet it often produces only partial progress rather than a fully connected W-clear passage from start to goal. Taken together, these comparisons support the need to couple frontier-gap-guided bottleneck reasoning with contact-feasible action generation.

The ablation study further clarifies which parts of the proposed formulation support this performance. Removing `presearch` reduces success from 92.5% to 85.0% and increases the number of simulations, suggesting that frontier-gap-guided prioritization helps reduce unnecessary validation and focus computation on useful bottlenecks. Removing `ModeTable` reduces success further to 80.0%, again with higher planning cost and more simulation calls, indicating that structured reuse of contact modes improves planning efficiency while helping preserve executable action generation under the same task setting. Together, these ablations support both sides of the proposed interface: frontier-gap-guided bottleneck prioritization on the representation side, and structured reuse of contact-feasible action templates on the execution side.

B. Hardware Execution

To verify that the frontier-gap-guided interventions remain valid beyond simulation, we deploy the method on a real two-robot platform in cluttered corridor-construction tasks. Each trial is conducted in a 5×6 m workspace with 2 UGVs and 6 movable obstacles. The scene state is observed online through motion capture and used to update execution as well as the current WCCG throughout the task.

A representative execution is shown in Fig. 3. Starting from an initially blocked scene, the robots sequentially open frontier gaps until a W-clear corridor emerges from start to goal. As obstacles are displaced, the WCCG is rebuilt from the observed scene configuration, so the planner continues reasoning over the currently reachable bottlenecks rather than following a fixed open-loop obstacle-removal sequence. This is important in hardware because small motion deviations can invalidate a purely geometric plan, whereas the updated graph keeps the intervention logic aligned with the evolving scene.

Overall, the hardware experiments support both sides of the proposed interface. On the representation side, the updated WCCG continues to expose the relevant frontier gaps as the scene evolves. On the execution side, the validated gap-opening actions remain executable in the representative hardware runs. Together, these results show that frontier gaps remain a useful interface between clearance reasoning and contact-feasible multi-robot pushing in representative real-world executions.

IV. Conclusion and Outlook

We presented *frontier gaps* as a structured interface between width-aware clearance reasoning and contact-feasible multi-robot pushing. In the proposed formulation, the WCCG identifies the reachable bottlenecks that currently block progress, each selected gap is converted into a compact set of contact candidates, and physics-informed search validates only the most promising gap-opening interventions. This interface allows cluttered path clearing to be treated as a single planning problem that links bottleneck selection to executable multi-robot action generation.

The current system still depends on sufficiently accurate state estimation and remains sensitive to drift, wheel slippage, and contact variability during real execution. Future work will extend this formulation to settings with limited visibility, uncertain object masses, and tighter coupling between estimation, replanning, and contact execution.

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